Cambridge (CIE) A Level Maths: Pure 1

Your notes

Inequalities & Simultaneous Equations

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Your notes

Linear simultaneous equations using elimination

What are linear simultaneous equations?

- When you have more than one equation in more than one unknown, then you are dealing with **simultaneous equations**
- An equation is **linear** if none of the unknowns in it is raised to a power other than one
- Solving a **pair of simultaneous equations** means finding **pairs of values** that make **both equations true** at the same time
- A linear equation in two unknowns will produce a straight line if you graph it... linear = line
- A pair of simultaneous equations will produce lines that will cross each other (if there is a solution!)

How do I use elimination to solve linear simultaneous equations?

Step 1: Multiply one (or both) of the equations by a constant (or constants) to get the numbers in front of **one** of the unknowns to match

Step 2: If the matching numbers have the **same** sign, then **subtract** one equation from the other. If the matching numbers have **different** signs then **add** the equations together

Step 3: Solve the new equation from Step 2 to find the value of one of the unknowns

Step 4: Substitute the value from Step 3 into one of the original equations, and solve to find the value of the other unknown

Step 5: Check your solution by substituting the values for the two unknowns into the original equation you **didn't** use in Step 4

Examiner Tips and Tricks

- Don't skip the checking step (it only takes a few seconds) – there are many places to go wrong when solving simultaneous equations!
- Mishandling minus signs is probably the single biggest cause of student error in simultaneous equations questions

Worked Example

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Your notes

Solve the simultaneous equations:

$$-2x + 4y = 5$$
$$4x - 5y = -7$$

$$-2x + 4y = 5 \quad (1)$$
$$4x - 5y = -7 \quad (2)$$

LABEL THE EQUATIONS

STEP 1: [2 × (1)] $-4x + 8y = 10$
[(2)] $4x - 5y = -7$

WE ONLY HAVE TO MULTIPLY ONE OF THE EQUATIONS HERE

STEP 2:
$$\begin{array}{r} -4x + 8y = 10 \\ +\ (4x - 5y = -7) \\ \hline 3y = 3 \end{array}$$

THE SIGNS IN FRONT OF THE 4's ARE DIFFERENT, SO ADD

STEP 3: $y = 3 \div 3 = 1$

STEP 4: $4x - 5(1) = -7$
$4x = -2$
$x = -0.5$

USING EQUATION (2)

STEP 5: $-2(-0.5) + 4(1) = 5$
$1 + 4 = 5$
$5 = 5$ ✓

USING EQUATION (1)

$x = -0.5 \quad y = 1$

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Linear simultaneous equations using substitution

What are linear simultaneous equations?

- When you have more than one equation in more than one unknown, then you are dealing with **simultaneous equations**
- An equation is **linear** if none of the unknowns in it is raised to a power other than one
- Solving a **pair of simultaneous equations** means finding **pairs of values** that make **both equations true** at the same time
- A linear equation in two unknowns will produce a straight line if you graph it... linear = line
- A pair of simultaneous equations will produce lines that will cross each other (if there is a solution!)

How do I use substitution to solve linear simultaneous equations?

Step 1: Rearrange one of the equations to make one of the unknowns the subject (if one of the equations is already in this form you can skip to Step 2)

Step 2: Substitute the expression found in Step 1 into the equation not used in Step 1

Step 3: Solve the new equation from Step 2 to find the value of one of the unknowns

Step 4: Substitute the value from Step 3 into the rearranged equation from Step 1 to find the value of the other unknown

Step 5: Check your solution by substituting the values for the two unknowns into the original equation you **didn't** rearrange in Step 1

Examiner Tips and Tricks

- Although elimination will always work to solve simultaneous linear equations, sometimes substitution can be easier and quicker.
- Knowing both methods can help you a lot in the exam (plus you will need substitution to solve quadratic simultaneous equations).

Worked Example

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Solve the simultaneous equations:

$$3x + 5y = -9$$
$$y = 1 - 9x$$

LABEL THE EQUATIONS

$3x + 5y = -9$ ①

$y = 1 - 9x$ ②

y IS ALREADY THE SUBJECT OF EQUATION ②, SO WE CAN SKIP STEP 1

STEP 2: $3x + 5(1 - 9x) = -9$

$3x + 5 - 45x = -9$

$-42x = -14$

SUBSTITUTE EQUATION ② INTO EQUATION ①

STEP 3: $x = -14 \div (-42) = \frac{1}{3}$

STEP 4: $y = 1 - 9(\frac{1}{3}) = 1 - 3 = -2$

USING EQUATION ②

STEP 5: $3(\frac{1}{3}) + 5(-2) = -9$

$1 - 10 = -9$

$-9 = -9$ ✓

USING EQUATION ①

$x = \frac{1}{3}$ $y = -2$

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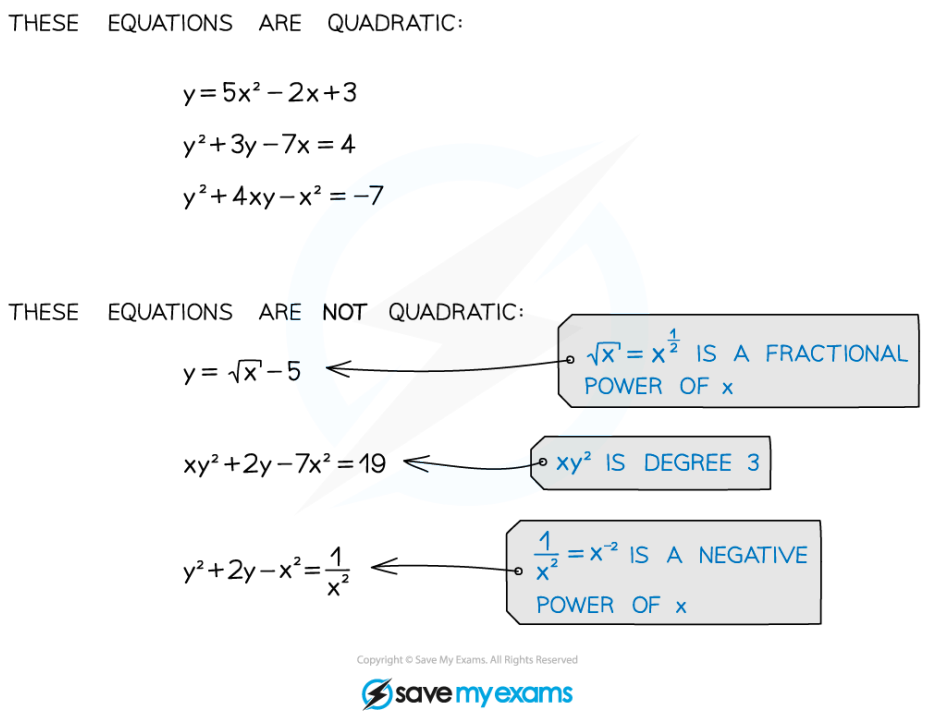
Quadratic simultaneous equations

What are quadratic simultaneous equations?

- When you have more than one equation in more than one unknown, then you are dealing with **simultaneous equations**
- An equation is quadratic if it contains terms of degree two, but no terms of any higher degrees (and also no unknowns raised to negative or fractional powers)



- Solving two simultaneous equations in two unknowns means finding pairs of values that make both of the equations true at the same time
  - At A level usually only one equation will be quadratic and the other will be linear
  - For one quadratic and one linear equation there will usually be **two** solution pairs (although there can be one, or none)

How do I solve quadratic simultaneous equations?

Step 1: Rearrange the **linear** equation so that one of the unknowns becomes the subject (if the linear equation is already in this form, you can skip to Step 2)

Step 2: Substitute the expression found in Step 1 into the quadratic equation

Step 3: Solve the new quadratic equation from Step 2 to find the values of the unknown (there will usually be two of these)

Step 4: Substitute the values from Step 3 into the rearranged equation from Step 1 to find the values of the other unknown

Step 5: Check your solutions by substituting the values for the two unknowns (one pair at a time!) into the original quadratic equation

Your notes

Examiner Tips and Tricks

- You have to use substitution to solve quadratic simultaneous equations – the elimination method won't work.

Worked Example

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Solve the simultaneous equations:

$$2x - 3y = 23$$
$$3y^2 = 4x^2 + 11$$

$2x - 3y = 23$ ← LINEAR

$3y^2 = 4x^2 + 11$ ← QUADRATIC

STEP 1: $2x - 3y = 23$

$2x = 3y + 23$ ← REARRANGE THE LINEAR EQUATION

BECAUSE THE QUADRATIC HAS $4x^2 = (2x)^2$ IT'S EASIEST TO STOP AT $2x$ HERE

STEP 2: $3y^2 = 4x^2 + 11$ ← SUBSTITUTE INTO THE QUADRATIC

$3y^2 = (2x)^2 + 11$

$3y^2 = (3y + 23)^2 + 11$

$3y^2 = 9y^2 + 138y + 529 + 11$

$6y^2 + 138y + 540 = 0$

[÷6] $y^2 + 23y + 90 = 0$

STEP 3: $(y+5)(y+18) = 0$ ← SOLVE BY FACTORISING (YOU COULD ALSO USE THE QUADRATIC FORMULA)

$y = -5$

OR $y = -18$

STEP 4: IF $y = -5$

$2x = 3(-5) + 23 = 8$ ← SUBSTITUTE INTO REARRANGED LINEAR EQUATION

$x = 4$

IF $y = -18$

$2x = 3(-18) + 23 = -31$

$x = -15.5$

STEP 5: $3(-5)^2 = 4(4)^2 + 11$ ← CHECK BY SUBSTITUTING INTO QUADRATIC!

$75 = 75$ ✓

$3(-18)^2 = 4(-15.5)^2 + 11$

$972 = 972$ ✓

THE SOLUTIONS ARE:

$x = 4$, $y = -5$

AND

$x = -15.5$, $y = -18$ ← INDICATE WHICH x GOES WITH WHICH y

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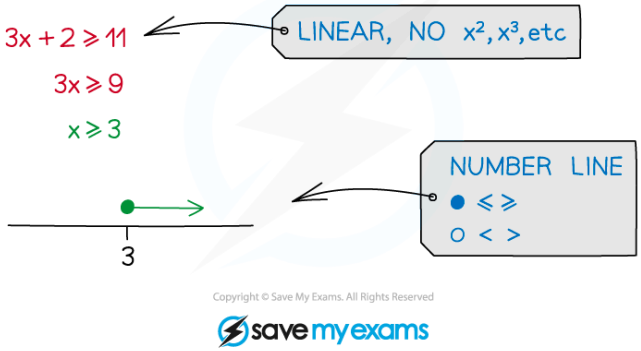
Linear inequalities

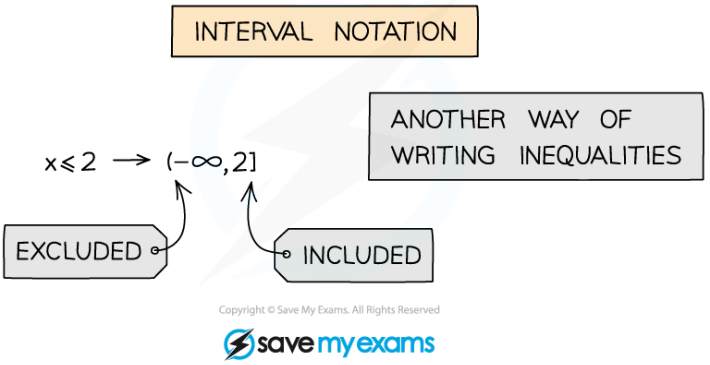
What are linear inequalities?

- Similar to equations but answers take a range of values
- Linear means there will be no terms other than degree 1
  - no squared terms or higher powers, no fractional or negative powers
- Number line diagrams are often used



What is interval notation?

- Use of square [] and round () brackets
- [or] mean included
- (or) mean excluded
  - (4,8] means $4 < x \leq 8$
- Note ∞ always uses (or)



Your notes

How do I solve linear inequalities?

- treat as equations but
  - avoid multiplying or dividing by a negative
  - if unavoidable, "flip" the inequality sign so $< \rightarrow >$, $\geq \rightarrow \leq$, etc
  - do rearrange to make the x term positive

$$8 - 3x \geqslant 5x - 4$$

AVOID MULTIPLYING OR DIVIDING BY A NEGATIVE

$$8 + 4 \geqslant 5x + 3x$$

REARRANGE TO MAKE x POSITIVE

$$8x \leqslant 12$$

$$x \leqslant \frac{3}{2}$$

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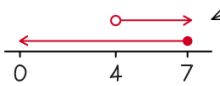
Worked Example

Solve the simultaneous inequalities 19-2x≥5 and 9<x+5.

$$19 - 2x \geqslant 5$$
$$19 - 5 \geqslant 2x$$
$$2x \leqslant 14$$
$$x \leqslant 7$$
$$9 < x + 5$$
$$x > 4$$

SOLVE EACH INEQUALITY SEPARATELY, DEALING WITH NEGATIVES CAREFULLY

DRAW A QUICK DIAGRAM TO SEE WHERE ANSWERS COINCIDE (OVERLAP)



$$4 < x \leqslant 7$$

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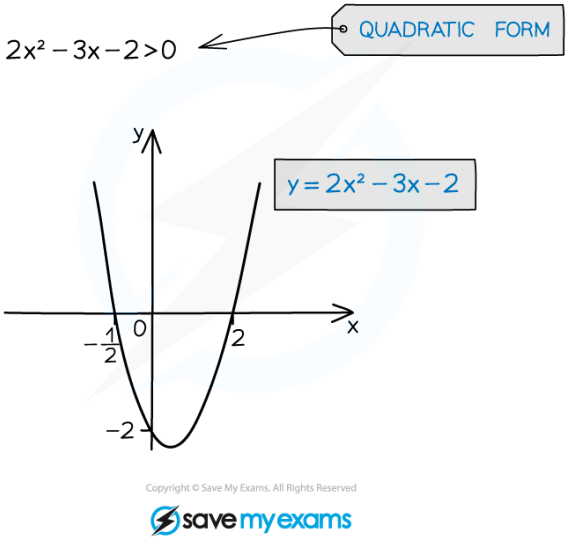
Quadratic inequalities

What is a quadratic inequality?

- Similar to quadratic equations
- Sketching a quadratic graph is essential
- Can involve the **discriminant** or **applications** in **mechanics** and **statistics**



How do I solve quadratic inequalities?

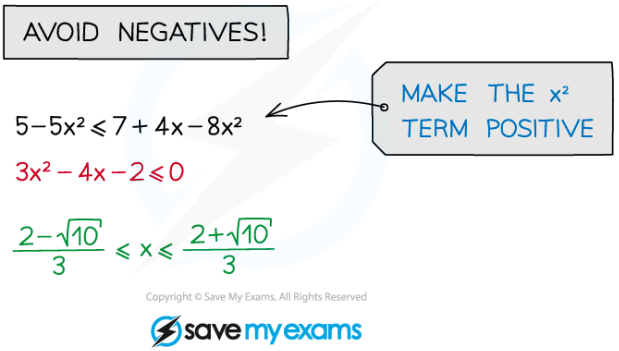
- **STEP 1**: **Rearrange** the inequality into quadratic form with a **positive squared term**
  - $\boldsymbol{ax^2 + bx + c > 0}$ (>, <, ≤ or ≥)
- **STEP 2**: Find the **roots** of the quadratic equation
  - **Solve** $ax^2 + bx + c = 0$ to get $\boldsymbol{x_1}$ and $\boldsymbol{x_2}$ where $\boldsymbol{x_1 < x_2}$
- **STEP 3**: **Sketch** a graph of the quadratic and **label the roots**
  - As the squared term is positive it will be **"U" shaped**
- **STEP 4**: **Identify** the **region** that satisfies the inequality
  - For $\boldsymbol{ax^2 + bx + c > 0}$ you want the region **above** the x-axis
    - The solution is $\boldsymbol{x < x_1}$ **or** $\boldsymbol{x > x_2}$
  - For $\boldsymbol{ax^2 + bx + c < 0}$ you want the region below the x-axis

- The solution is $x > x_1$ **and** $x < x_2$
- This is more commonly written as $x_1 < x < x_2$

- Be careful:
  - **avoid** multiplying or dividing by a negative number

    if unavoidable, "**flip**" the inequality sign so $< \rightarrow >$, $\geq \rightarrow \leq$, etc
  - **avoid** multiplying or dividing by a **variable** (**x**) that **could** be negative

    (multiplying or dividing by $\mathbf{x^2}$ guarantees positivity (unless **x** could be 0) but this can create extra, invalid solutions)
  - **do** rearrange to make the x^2 term positive



Your notes

Worked Example

Find the set of values for which $3x^2 + 2x - 6 > x^2 + 4x - 2$.

$3x^2 + 2x - 6 - x^2 - 4x + 2 > 0$

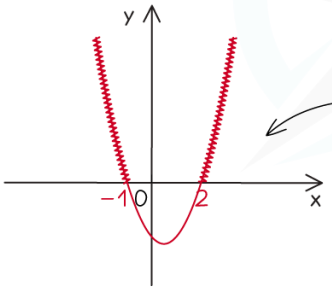
$2x^2 - 2x - 4 > 0$

REARRANGE TO QUADRATIC FORM

$2(x^2 - x - 2) > 0$

$x^2 - x - 2 > 0$

$(x - 2)(x + 1) > 0$



CRUCIAL: SKETCH THE GRAPH TO SEE WHERE THE SOLUTIONS ARE

$x < -1$ OR $x > 2$

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Examiner Tips and Tricks

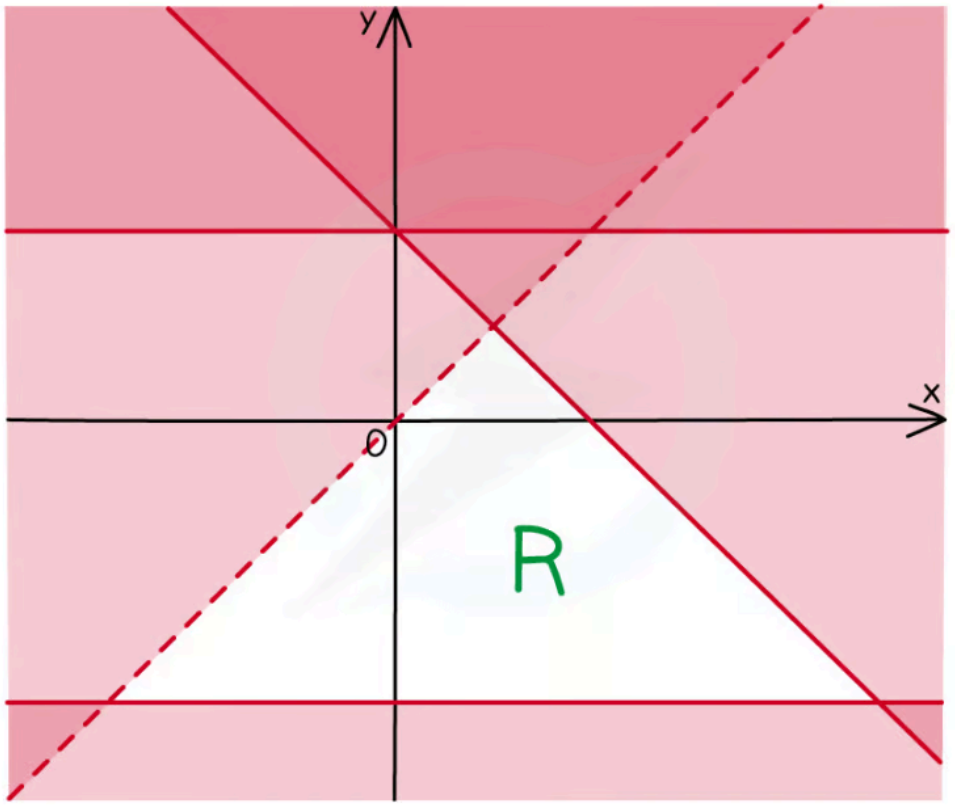
- A calculator can be super-efficient but some marks are for method.
- Use your judgement:
  - is it a "show that" or "prove" question?
  - how many marks?
  - how long is the question?

Your notes

Inequalities on graphs

What are inequalities on graphs?

- Inequalities can be represented on graphs by shaded regions and dotted or solid lines



- These inequalities have two variables, **x** and **y**
- Several inequalities are used at once
- The solution is an **area** on a graph (often called a **region**)
- The inequalities can be **linear** or **quadratic**

How do I draw inequalities on a graph?

- Sketch each graph
  - If the inequality is **strict (< or >)** then use a **dotted line**
  - If the inequality is **weak (≤ or ≥)** then use a **solid line**
- Decide **which side** of the line **satisfies** the inequality
  - Choose a coordinate on each side and **test** it in the inequality

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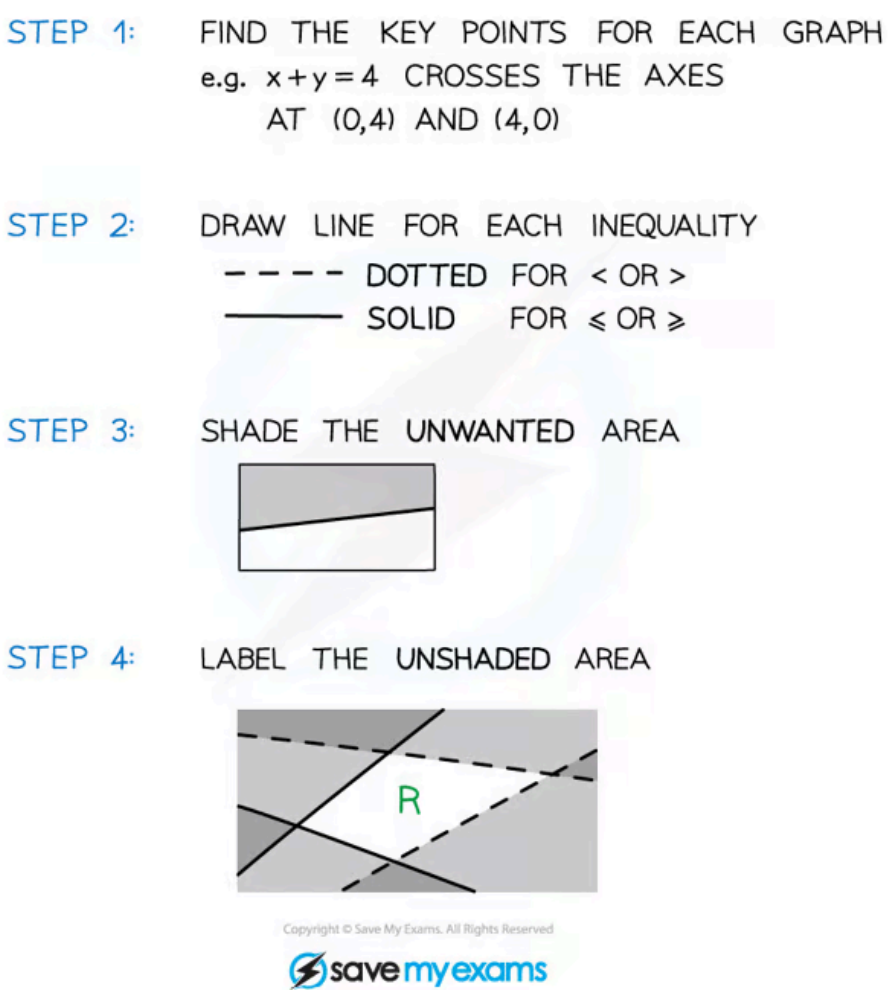
- The origin is an easy point to use
- If it satisfies the inequality then that whole side of the line satisfies the inequality
  - For example: (0,0) satisfies the inequality $y < x^2 + 1$ so you want the side of the curve that contains the origin



Examiner Tips and Tricks

- **Recognise** this type of inequality by the use of **two** variables
- You may have to **deduce** the inequalities from a **given** graph
- Pay careful attention to which **region** you are **asked to shade**
- Sometimes the exam could ask you to shade the region that **satisfies the inequalities this means** you should **shade** the region that is **wanted**.
  - If you're unsure, you could ...
  - ... draw the (dotted and/or solid) lines in on the answer diagram and use a rough sketch to find the region required ...
  - ... write clearly you have "shaded the unwanted area"
  - ... and/or ...

Your notes

- ... write clearly you have "shaded the unwanted area
- As long as your final answer is clear you should get the marks!

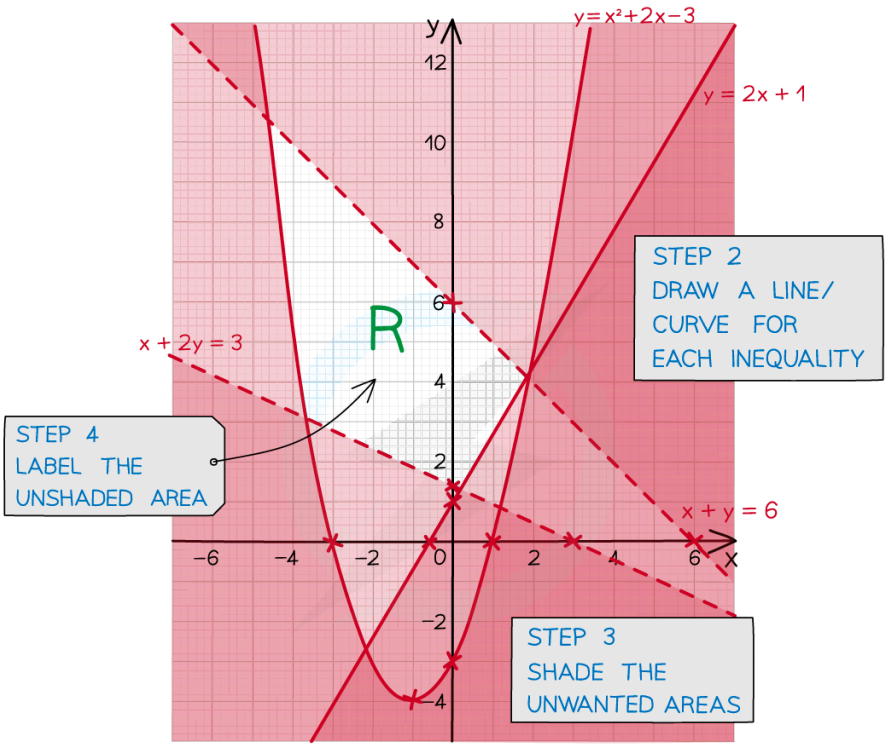
Your notes

Worked Example

Your notes

On the graph below show the region bounded by the following inequalities:

$$y \geq x^2 + 2x - 3 \qquad x + y < 6$$
$$y \geq 2x + 1 \qquad x + 2y > 3$$



TURNING POINT

$y = x^2 + 2x - 3 = (x+3)(x-1)$

$(0,-3)$ $(-3,0)$ $(1,0)$ $(-1,-4)$ "$\geqslant$ SOLID"

$x + y = 6$

$(0,6)$ $(6,0)$ "<DOTTED"

$y = 2x + 1$

$(0,1)$ $(\frac{-1}{2},0)$ "$\geqslant$ SOLID"

$x + 2y = 3$

$(0,\frac{3}{2})$ $(3,0)$ ">DOTTED"

STEP 1
FIND THE KEY POINTS
FOR EACH GRAPH

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